

Fuel energy efficiency and rebound effect in in European road transport: a semiparametric stochastic frontier analysis

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Abstract

Improving fuel consumption efficiency is a key objective in the European Union's strategy for achieving climate neutrality and reducing dependency on fossil fuels. This study investigates the fuel efficiency performance of European countries over the past decade using a semiparametric stochastic frontier analysis. The study draws on a balanced panel dataset comprising 27 EU member states and selected neighboring countries from 2014 to 2023. Results highlight significant disparities in fuel efficiency across countries and suggest that structural factors—such as fuel prices, length of roads and . . . play a critical role in shaping national performance. The findings provide useful insights for policymakers aiming to target inefficiencies and design more effective energy and environmental interventions.

Keywords:

JEL Classifications:

1. Introduction

The transport sector constitutes a fundamental component of global economic development, underpinning productivity, urbanization, and trade ([Rodrigue, 2020](#)). Beyond facilitating the movement of goods and people, transport infrastructure enhances spatial connectivity and promotes economic integration and social cohesion. Among transport modes, road transport remains dominant due to its flexibility, accessibility, and extensive coverage. Historically, its expansion has shaped economic and social development, both driving and responding to industrialization and globalization processes.

The role of public infrastructure investment, particularly in transport, has been a central theme in the economic growth literature since the seminal contribution of ([Aschauer, 1989](#)), who established a positive link between public capital and private sector productivity. Subsequent studies (e.g. [Gramlich, 1994](#)) confirmed that efficient transport infrastructure reduces transaction costs, facilitates trade, and enhances competitiveness.

Nevertheless, the continuous expansion of road transport for freight and passenger activities has generated increasing energy demand and adverse environmental externalities ([Schipper et al., 2011](#); [Creutzig et al., 2015](#)). The sector accounts for a substantial share of global petroleum consumption and greenhouse gas (GHG) emissions, contributing to air pollution,

climate change, and energy insecurity. Consequently, improving fuel efficiency in road transport has become an economic and environmental imperative.

Within Europe, energy efficiency has been established as a central pillar of climate and energy policy. The European Green Deal and the EU’s target of climate neutrality by 2050 highlight efficiency improvements as a cost-effective strategy for reducing GHG emissions, strengthening energy security, and fostering sustainable growth (European Commission, 2020). Despite these efforts, progress in decarbonizing the transport sector has been limited. The EEA (2024) reports that while emissions from domestic aviation, navigation, and rail have declined since 1990, road transport still represents about 73% of total EU transport GHG emissions, with only marginal reductions in recent years (−0.8% in 2023 relative to 2022).

In this context, understanding the determinants of fuel consumption is essential for designing effective fiscal, environmental, and transport policies. Fuel demand in the road sector encapsulates both behavioral responses and structural characteristics, influencing energy security, emissions trajectories, and macroeconomic performance. The empirical literature on fuel demand elasticities (Goodwin et al., 2004; Dahl, 2012) emphasizes that price and income elasticities play a critical role in policy evaluation especially in assessing the role of fuel taxation, carbon pricing, and efficiency standards. Moreover, cross-country heterogeneity in vehicle ownership, infrastructure, and economic structure warrants explicit econometric modeling to capture structural and policy-driven effects.

At the macroeconomic level, fuel efficiency can be interpreted as the capacity of a country to generate economic output while minimizing fuel input. However, cross-country comparisons are inherently complex due to structural heterogeneity in income levels, industrial composition, geographic conditions, and policy regimes. Conventional indicators—such as fuel intensity (fuel per unit of GDP)—ignore these heterogeneities and may yield biased inferences regarding relative performance (Zhou et al., 2008).

To overcome these limitations, the econometric framework developed to estimate energy efficiency and corresponding rebound effect is based on stochastic frontier analysis (SFA, Kumbhakar and Lovell, 2000). Boyd (2008) is an early example of research estimating an energy input requirement function using SFA to analyze the efficient use of energy in the industrial sector. Although several studies have applied SFA to evaluate energy and environmental efficiency, relatively few have focused specifically on fuel consumption efficiency at the country level within Europe. Existing works tend to emphasize broader energy efficiency or carbon emissions efficiency (Filippini and Hunt, 2011; Barros et al., 2013), while studies focusing explicitly on fossil fuel use often restrict their scope to the firm or sectoral level. This paper fills a gap in the literature by applying SFA to a panel dataset of European countries to estimate their technical efficiency in fuel consumption over time.

The concept of frontier efficiency, which measures inefficiency as the deviation of a unit from the efficient or best practices frontier, requires to start with a production function estimation. However, the choice of an appropriate functional form is a non trivial issue for production studies and model builder’s task becomes tricky due to the different available functional forms (Griffin et al., 1987). The topic is typically overlooked in applications and the canonical functions are commonly used (i.e. Cobb-Douglas or Translog). As also stated in the seminal work of Meeusen and van den Broeck (1977), any specification of the production function is debatable and it is useful to implement a sound sensitivity analysis with respect to the particular choice. Especially in the context of energy analysis, Saunders (2000) underline that “The choice of production function in theoretical and empirical studies of rebound

matters to the conclusions”. Finally, [Filippini and Orea \(2014\)](#) proposed the use of semi-parametric stochastic frontier models in disentangling transient and persistent inefficiency, particularly in the context of energy economics.

Building on a semiparametric SFA framework, the empirical model explicitly separates transient (time-varying) from persistent (time-invariant) inefficiency components. This decomposition enables a more nuanced interpretation of efficiency dynamics and policy effects over time. To the best of our knowledge, this is the first cross-country application integrating semiparametric frontier estimation with inefficiency decomposition for road transport fuel consumption.

This study aims to contribute to this literature by investigating the determinants and efficiency of fuel use in the road transport sector across 32 European countries over the period 2014–2023. The specific objectives are threefold:

- to estimate country-level fuel efficiency scores using a stochastic frontier approach that accounts for cross-sectional and temporal heterogeneity;
- to investigate the influence of economic, infrastructural, and demographic factors on fuel demand;
- to evaluate the presence of rebound effects in energy use across European economies.

The empirical model specifies fuel consumption as the dependent input variable, jointly determined with key production factors such as GDP and labor, and augmented by control variables including vehicle stock and diesel price, which proxy for technological, structural, and behavioral effects. Estimation relies on panel data techniques, incorporating country effects to capture unobserved heterogeneity and time effects to account for macroeconomic shocks and policy interventions. The empirical results provide new evidence on the efficiency of fuel utilization and contribute to ongoing debates on energy transition and transport decarbonization in the EU.

The remainder of the paper is organized as follows: Section 2 reviews relevant literature on fuel and energy efficiency, with a focus on SFA applications in macroeconomic settings. Section 3 describes the methodological framework while Section 4 presents the data sources, variable definitions. Section 5 reports the empirical results and discusses their policy implications. Finally, Section 5 concludes with a summary of key findings and directions for future research.

NOTES Why It Matters for European Transport Policy: Achieving Climate Targets: Rebounds, like increased driving from cheaper travel, can offset savings from electric vehicles or stricter efficiency standards, making climate goals harder to reach. Accurate Policy Evaluation: Ignoring rebounds risks overestimating benefits and investing in solutions that fail to deliver expected environmental improvements. Understanding Behavioral Changes: Efficiency reduces travel costs, prompting more vehicle miles traveled (VMT), increasing congestion, noise, and pollution even with cleaner vehicles. Freight Sector Impact: Efficiency gains can boost goods movement, increasing road wear and emissions—a key concern for Europe. Designing Better Solutions: Insights into rebound effects support “reboundless” policies, combining efficiency with cost internalization (e.g., road pricing, taxes) or modal shifts (walking, cycling, public transport) to genuinely reduce transport demand. Avoiding Misplaced Investments: Recognizing rebounds ensures strategies go beyond technology alone, targeting the root causes of transport demand. Bottom Line: Technological fixes alone aren’t

enough—understanding behavior is crucial to creating sustainable, effective transport systems in Europe.

2. Literature review

3. Methods

In the stochastic frontier framework, the input requirement frontier represents the minimum input required to produce a given level of output, given input prices and technology. The measurement of energy efficiency requires knowledge of the frontier function, which allows for the comparison of individual production levels with the efficient frontier of the possibilities set (Farrell, 1957).

The stochastic energy input frontier model for panel data can be written as:

$$\ln E_{it} = \alpha_0 + C(\mathbf{x}_{it}; \boldsymbol{\beta}) + v_{it} + u_{it}, \quad (1)$$

where E_{it} denotes the total amount of energy consumed by unit i at time t , $C(\cdot)$ defines the input requirement frontier relationship between the single input and the output/inputs depending on the corresponding parameter $\boldsymbol{\beta}$ vector (Filippini and Hunt, 2015). The composite error term consists of two components $\varepsilon_{it} = v_{it} + u_i$, where $v_{it} \sim N(0, \sigma_v^2)$ is a symmetric random error capturing statistical noise and measurement errors, while $u_i \geq 0$ represents the inefficiency component specific to each unit, that is, the degree to which each observation's cost exceeds the minimum attainable input frontier.

When panel data are available, more information can be extracted about individual units, and a key advantage of this framework is the ability to account for unobserved heterogeneity, which is not possible in cross-sectional approaches. The specification and estimation of true fixed- or random-effect stochastic frontier models has been the focus of recent research, with the main objective of obtaining estimates of persistent inefficiencies.

Building on standard panel data models, where the focus is on controlling for unit-specific effects due to unobserved, time-invariant factors, this approach extends the early panel stochastic frontier models of Pitt and Lee (1981) and Schmidt and Sickles (1984). Several extensions have been developed based on different assumptions: some treat all time-invariant effects (fixed or random) as persistent inefficiency, while others separate unit-specific effects from inefficiency. In the stochastic frontier literature, time-invariant heterogeneity and persistent inefficiency have historically been treated differently. Early attempts to incorporate persistent inefficiency include Kumbhakar and Heshmati (1995) who distinguished between persistent and time-varying inefficiency, but did not disentangle persistent inefficiency from unit-specific effects while Wang and Ho (2010) explicitly account for unobserved time-invariant heterogeneity, but do not consider persistent inefficiency.

Kumbhakar et al. (2014) and Colombi et al. (2014) introduced a new approach that account simultaneously for persistent and transient inefficiency along with unit-specific effects. They argue that failing to separate persistent inefficiency from time-invariant heterogeneity leads to model misspecification and biased efficiency estimates. In their specification, the model error is decomposed into four components:

$$E_{it} = \alpha_0 + C(\mathbf{x}_{it}; \boldsymbol{\beta}) + \mu_i + v_{it} + \eta_i + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (2)$$

where μ_i captures unit-specific latent heterogeneity or the unobserved effect of time-invariant inputs, v_{it} represents random shocks, η_i captures persistent (time-invariant) inefficiency, and u_{it} captures transient (time-varying) inefficiency. In the presence of long-term structural issues and systematic shortfalls, persistent inefficiency may be confounded with unit-specific effects. Ignoring this persistent inefficiency can lead to an upward bias in the estimation of total efficiency.

Due to the complexity of the model, various estimation methods have been proposed. For instance, [Colombi et al. \(2014\)](#) employed a full maximum likelihood approach, [Tsionas and Kumbhakar \(2014\)](#) adopted a Bayesian approach and [Filippini and Greene \(2016\)](#) applied a simulated maximum likelihood (SML) method. For the model estimation we consider the simple multi-step procedure adopting the approach given in [Kumbhakar et al. \(2014\)](#) where equation (2) is rewritten as:

$$y_{it} = \alpha_0^* + C(\mathbf{x}_{it}; \boldsymbol{\beta}) + \alpha_i + \epsilon_{it}, \quad (3)$$

where $\alpha_0^* = \alpha_0 - E(\eta_i) - E(u_{it})$, $\alpha_i = \mu_i + \eta_i + E(\eta_i)$ and $\epsilon_{it} = v_{it} + u_{it} + E(u_{it})$. More specifically, given that α_i and ϵ_{it} have zero mean and constant variance, this model can be estimated in three steps:

1. equation (3) corresponds to the standard regression based panel data model, consequently the standard fixed/random effects panel model is estimated obtaining an estimate of $\boldsymbol{\beta}$ and predicted values of α_i and ϵ_{it} , denoted as $\hat{\alpha}_i$ and $\hat{\epsilon}_{it}$
2. the time-varying technical inefficiency u_{it} is estimated by considering a stochastic frontier model on the predicted value $\hat{\epsilon}_{it}$ from step 1 under standard distributional assumptions, obtaining a prediction of the country's time-varying technical inefficiency $\hat{u}_{it}$ using [Jondrow et al. \(1982\)](#) estimator $TE_{it} = \exp(-\hat{u}_{it})$
3. persistent inefficiency η_i is estimated by considering a stochastic frontier model on the predicted value $\hat{\alpha}_i$ from step 1 under standard distributional assumptions and the value of persistent technical efficiency is found by $PE_i = \exp(\hat{\eta}_i)$ where $\hat{\eta}_i$ is (again) the [Jondrow et al. \(1982\)](#) estimator.

Overall technical efficiency (OE) is obtained by the product of persistent and transient inefficiency.

A key challenge in estimating model (3) lies in specifying the functional form of the frontier $C(\cdot)$, which is generally unknown in practice. Empirical studies often rely on standard parametric forms such as the Cobb-Douglas or the Translog specifications, or other ad hoc formulations. However, when there is no strong theoretical or empirical justification for a particular specification, conducting sensitivity analysis becomes essential ([Giannakas et al., 2003](#)). Unlike the distributional assumptions for the error terms, the choice of the functional form $C(\cdot)$ is particularly critical. Even if the composed error terms are correctly specified, a misspecified frontier function can lead to biased estimates, particularly in the identification of influential input factors and in the calculation of technical efficiency ([Fan et al., 1996](#)). While the primary objective of model (2) is the estimation of the stochastic frontier, the derived technical efficiency scores - which measure the distance of observed input use from the estimated frontier - are also crucial, especially for evaluating energy efficiency.

In the classical stochastic frontier analysis (SFA) framework introduced by (Aigner, Lovell and Schmidt, 1977; Meeusen and van den Broeck, 1977) the frontier function is typically assumed to be linear in parameters. To relax this restrictive assumption, several alternative approaches have been proposed. Giannakas et al. (2003), for example, suggest applying Box-Cox transformations to both the dependent and independent variables to introduce functional flexibility. Nonparametric methods have also been explored; Fan et al. (1996) employ kernel smoothing techniques, while Kumbhakar et al. (2007) propose a local maximum likelihood approach that also allows flexibility in specifying the distribution of the error terms. However, these methods are not without limitations. The computational demands can become substantial, even when only a few input variables are involved, particularly in the case of local maximum likelihood estimation which is generally not recommended as a smoothing technique (pag 169 Hastie and Tibshirani, 1990). Furthermore, both nonparametric approaches are subject to the ‘curse of dimensionality’: as the number of input variables increases, the performance of the frontier estimator deteriorates significantly, and estimation becomes computationally intensive.

More recently, Ferrara and Vidoli (2017) suggested a Generalized Additive Model (GAM) framework where, starting from the work of Fan et al. (1996), the conditional expectation of the response is specified by a typical GAM in the form:

$$C(\mathbf{x}_{it}; \boldsymbol{\beta}) = \psi_0 + \sum_{j=1}^p \psi_j(x_j), \quad (4)$$

where each $\psi_j(\cdot)$ is a smooth function of the input x_j estimated from data, standardized so that $E[\psi_j(x_j)] = 0$ (Hastie and Tibshirani, 1990). The variability of the response is modeled through an additive function of the input variables, allowing for non-linear relationships and potentially improving the accuracy of the predictions. Because each additive component is estimated using a univariate smoother, the approach effectively avoids the *curse of dimensionality*. Moreover, the optimal transformations of the inputs are determined simultaneously during the estimation process.

This class of model includes the linear model as a special case, where $\psi_j(x_j) = \beta_j x_j$ (Aigner, Lovell and Schmidt, 1977) but it is clearly more general, because the ψ_j ’s can be very arbitrary nonlinear functions. In addition, it is still possible to further customize the specification of the production frontier considering a generalization of equation (4) in a such way

$$C(\mathbf{x}_{it}; \boldsymbol{\beta}) = \psi_0 + \sum_{j=1}^p \psi_j(x_j) + \sum_{i=1}^p \sum_{k < p} \psi_{kj}(x_k, x_j) + \beta z + \dots,$$

by introducing effects due to interactions among covariates or linear terms.

Given these premises, the stochastic frontier model (3) is estimated by replacing, in the first phase of the multi-step procedure, the conventional fixed or random effects specification with the estimation of the conditional expectation modeled via a Generalized Additive Model (4) with country-specific random intercepts¹ while the corrected ordinary least squares (COLS), obtained from the second and third moments of the OLS residuals (Kumbhakar and Lovell,

¹All the analyses presented in this paper have been carried out using the R Core Team (2024).

2000), for the error term parameters in steps 2-3. To flexibly model non-linear relationships ψ'_j s we employ penalized splines (P-splines Eilers and Marx, 1996). P-splines use a B-spline basis combined with a difference penalty to control the smoothness of the estimated function. This approach offers several methodological advantages over alternative smoothing techniques such as cubic splines, LOESS, and kernel smoothers. In particular, P-splines avoid the need for knot selection, provide stable and computationally efficient estimation, and mitigate the curse of dimensionality in additive models by applying univariate smoothers (Marx and Eilers, 2018). Their flexibility and ease of integration into generalized additive models make them especially suitable for high-dimensional or complex functional forms (Wood, 2017).

In order to control for the effect of technological change we introduce a set of time dummy variables instead of a time trend. Indeed, the latter is a more restrictive version to take into account for changes in exogenous impacts that cannot usually be measured explicitly, consistent with the idea of an underlying energy demand trend (Filippini and Lester, 2015).

As a result, our proposal incorporates a decomposition of efficiency into short-run and long-run components and adopts a semiparametric method for estimating the input requirement frontier, thereby addressing two critical aspects in the analysis of energy efficiency that are often overlooked in applied research as emphasized by Saunders (2000) and Filippini and Orea (2014).

After estimating energy efficiency, we calculate the rebound effect

$$R = 1 + \eta^E \quad (5)$$

where η^E is the elasticity of energy consumption with respect to energy efficiency. By decomposing overall efficiency into transient and persistent components, we are able to evaluate separately the two distinct sources of inefficiency. We estimate two distinct models to capture the short-term rebound effect (SRE) and the long-term rebound effect (LRE), rather than relying on a single unified specification (Adetutu et al., 2016). This framework allows for a clearer distinction between temporary behavioral adjustments and more structural, long-run responses to improvements in energy efficiency.

To address potential econometric issues associated with panel data estimation for the SRE, this study applies the Generalized Method of Moments (GMM) framework. The GMM estimator is particularly suited for dynamic panel models where regressors may be endogenous and unobserved heterogeneity is present. A key advantage of the GMM approach is its capacity to address endogeneity by using lagged values of endogenous or predetermined variables as internal instruments. This is especially important in the present context, since variables such as energy efficiency and fuel consumption may exhibit simultaneity or reverse causality. To this end, the study adopts the Arellano and Bond (1991) dynamic panel estimator, which specifies a first-differenced model to eliminate fixed effects and then applies valid moment conditions to ensure consistent estimation.

Two variants of the estimator are typically employed: the one-step and two-step GMM estimators. Both account for potential heteroskedasticity and autocorrelation in the idiosyncratic errors. The one-step estimator uses a weighting matrix independent of estimated parameters, while the two-step estimator applies an optimal weighting matrix derived from the covariance of the moment conditions. The latter is asymptotically more efficient, particularly in large samples and under non-i.i.d. disturbances. The two-step system GMM estimator is typically preferred, as it yields efficiency gains and employs a robust variance-covariance matrix obtained from the first-step estimates, ensuring consistency in the presence of panel-specific

heteroskedasticity and serial correlation (Arellano and Bond, 1991). In the empirical analysis we consider both estimator to better evaluate the robustness of the model specification. On the contrary, from a short-term standpoint, each country has a single efficiency score within each sector (i.e. persistent). For the SRE, we use a standard cross section regression model where the average energy fuel consumption over time serves as the response variable and persistent energy efficiency is treated as a regressor alongside the average values of the other variable over time (Adetutu et al., 2016).

4. Data and variables

The empirical analysis is based on a balanced panel data for 32 European countries over the period 2014–2023. The dataset was constructed using harmonized data retrieved from multiple internationally recognized statistical agencies, ensuring cross-country comparability and temporal consistency. The dependent variable in the input requirement function is annual fuel consumption, while explanatory variables capture economic, infrastructural, and demographic factors influencing fuel demand.

The primary data sources and variables are as follows:

- Fuel consumption by road vehicles, expressed in thousand tonnes of oil equivalent (Eurostat);²
- Number of passenger cars (Eurostat);
- Number of motor coaches, buses, and trolleybuses (Eurostat);
- Total length of roads (motorways, state roads, provincial roads, communal roads, and other roads), expressed in kilometers (UNECE Sustainable Transport Division);
- GDP per capita (constant 2015 US\$) (World Bank);
- Number of employed persons aged 15–64, expressed in thousands (Eurostat);
- Diesel price, expressed per 1,000 liters (European Commission).

The selection of variables in this study is grounded in standard economic and transport demand theory, which links fuel consumption to price, income, vehicle ownership, infrastructure, and demographic factors (Wadud et al., 2004). According to the law of demand, an increase in fuel prices is expected to reduce fuel consumption, as higher costs discourage driving and promote energy efficiency or alternative transport modes (Goodwin et al., 2004). Conversely, higher levels of income, measured by GDP per capita, generally lead to greater mobility and vehicle ownership, thereby increasing fuel consumption (Akinboade et al., 2010). The number of passenger cars and commercial vehicles (buses, coaches, trolleybuses) captures the size of the vehicle stock, a direct determinant of aggregate fuel demand, as more vehicles translate into higher potential fuel use (Goodwin et al., 2004). Road infrastructure, proxied by total road length, facilitates accessibility and reduces travel time, often inducing additional traffic

²Off-road use of fuels (for example: cranes and excavators at construction sites, harvesters and tractors on fields) is excluded from road transport and included in the respective sector of consumption (such as industry or agriculture).

and higher fuel use, a phenomenon known as induced demand (Duranton and Turner, 2011). Finally, the number of employed persons aged 15–64 reflects the intensity of economic activity and commuting demand, both of which are positively associated with fuel consumption. Together, these variables provide a comprehensive representation of the economic, structural, and demographic determinants of fuel demand in European road transport.

Table 1 reports the descriptive statistics for the key variables used in the analysis. The data reveal considerable heterogeneity across countries. Fuel consumption averages 9,137 thousand tonnes with a large standard deviation (12,667), while the number of passenger cars and motor coaches also exhibit substantial variability. The total length of roads shows wide dispersion, reflecting differing levels of infrastructure development. Economic indicators display similar variation, with GDP per capita averaging 32,453 and employment 7,049 thousand persons, whereas diesel prices are relatively stable (mean = 1.328, SD = 0.241). The boxplots in Figure

Table 1: Descriptive statistics.

Statistic	Mean	St. Dev.
Fuel consumption (thousands tonnes)	9,137	12,667
Number of passenger car (thousands)	8,105	12,078
Number of motor coaches, buses, and trolleybuses	44,973	119,706
Total length of roads (kilometers)	252,254	395,562
GDP per capita	32,453	23,455
Number of employed persons (thousands)	7,049	9,625
Diesel price	1.328	0.241

1 illustrate the temporal evolution of these variables from 2014 to 2022. Fuel consumption and road length show moderate fluctuations over time, while diesel prices exhibit a slight upward trend. The number of passenger cars and buses increases gradually, accompanied by widening interquartile ranges, suggesting growing disparities across regions. GDP per capita shows a consistent rise with reduced variability toward 2022, indicating economic convergence. Employment levels follow a similar upward trajectory with relatively stable dispersion.

Overall, the figures and descriptive statistics indicate significant cross-sectional differences but generally positive trends in transport activity and economic performance over the study period.

5. Results

5.1. Energy Efficiency from semiparametric SFA

Estimation results of model (3) under GAM specification are reported in Table 2 and the corresponding marginal effects described in Figure 2. The empirical estimates show that all regressors are statistically significant, confirming their relevance in explaining fuel consumption dynamics. The semiparametric specification further uncovers significant nonlinearities in the underlying relationships, indicating the presence of heterogeneity among countries. In particular, the estimated relationship between fuel consumption suggest that, at lower levels, an expansion in the number of buses contributes to reduced fuel consumption by promoting collective transport and reducing individual vehicle use. However, beyond a threshold, further

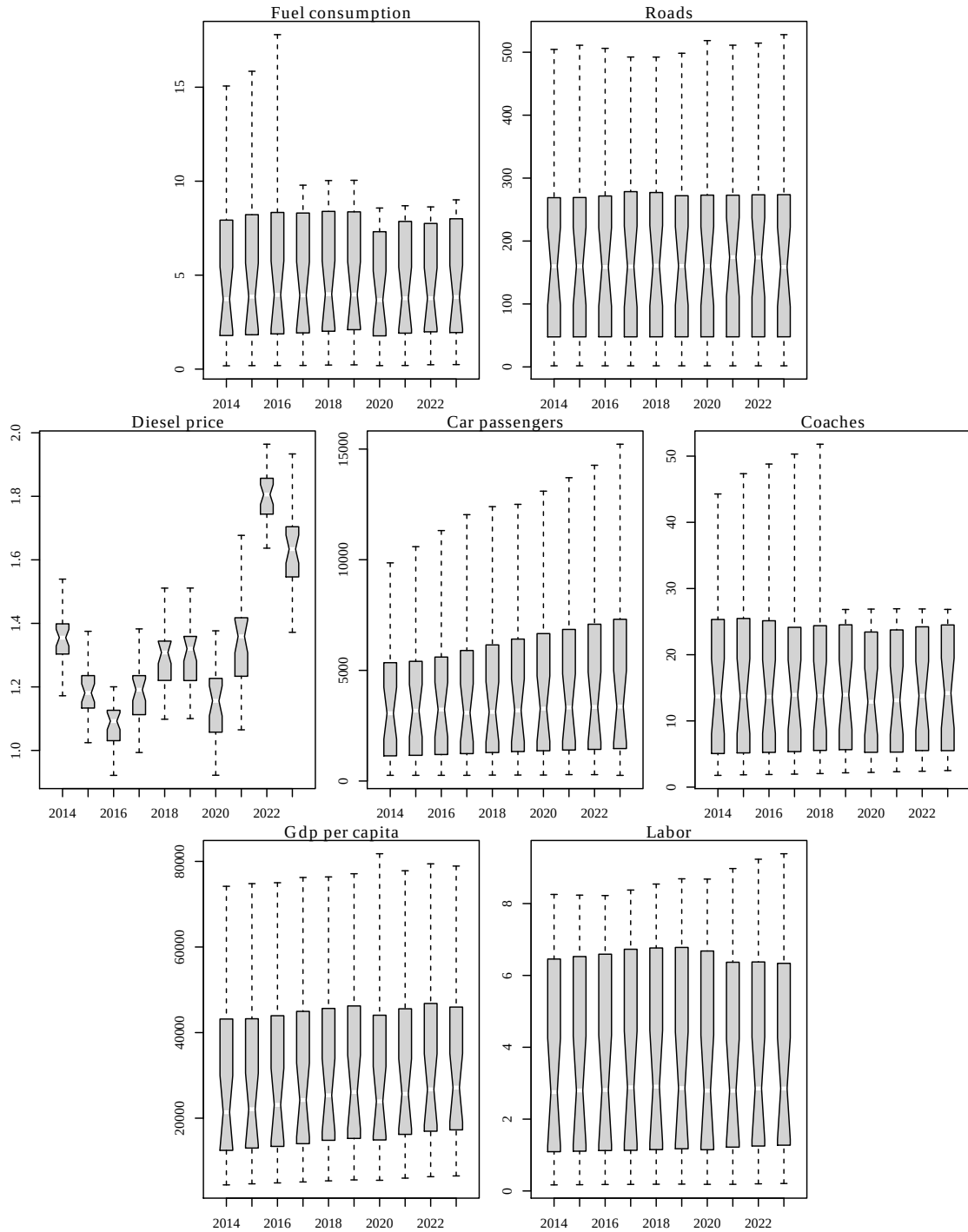


Figure 1: Annual distribution of selected variables.

Table 2: Parameter estimates.

	Estimate	Sd. Error	Pvalue
ps(roads)	0.072	0.002	***
ps(labor)	0.167	0.013	***
carpass	0.853	0.011	***
ps(coaches)	-0.095	0.005	***
gdp	0.270	0.004	***
ps(diesel price)	-0.686	0.032	***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1			

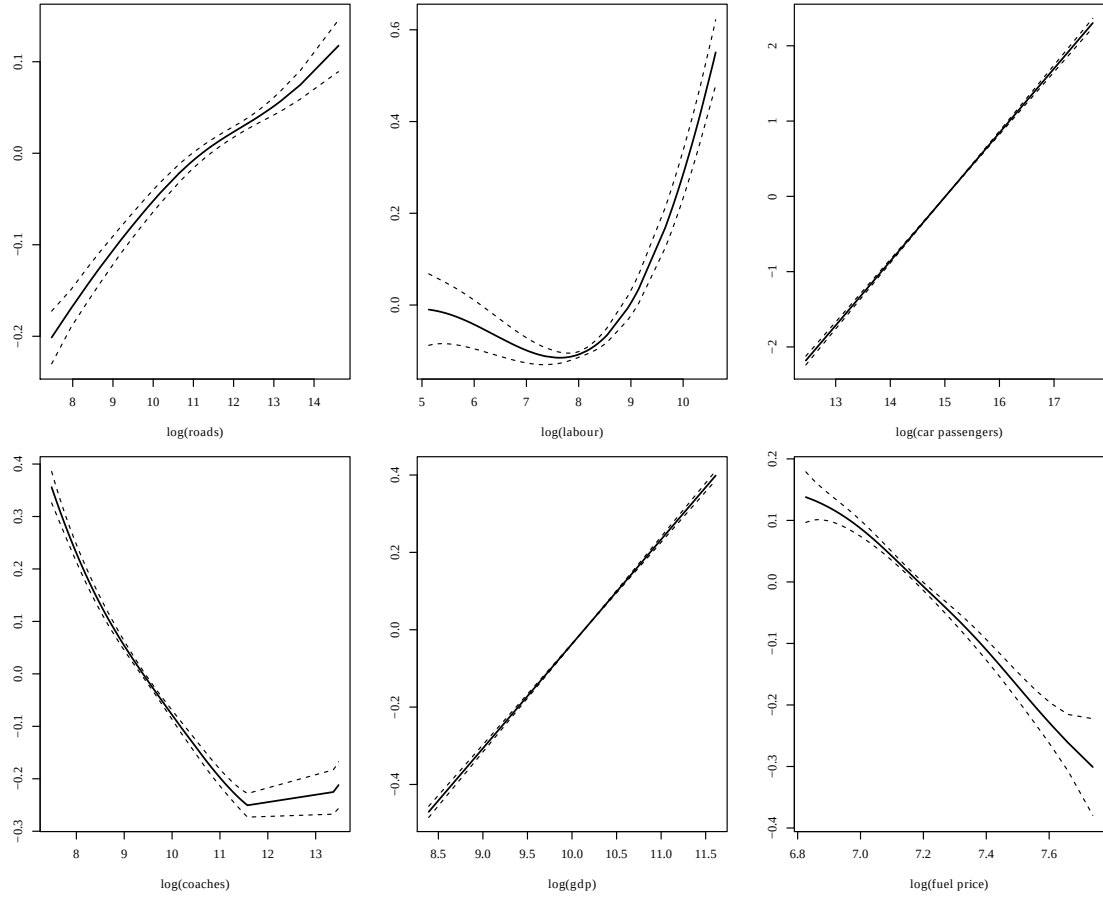


Figure 2: Partial effects estimated by GAMs.

increases in the bus fleet appear to generate the opposite effect, potentially due to network congestion or underutilized capacity. As expected from the relevant literature ([Goodwin et al., 2004](#); [Kouris, 1983](#)), there is a positive association between fuel consumption and the number of passenger cars, the total length of roads, the GDP per capita and the number of employed persons while a negative relationship with diesel price.

These nonlinear dynamics underscore the importance of considering heterogeneous marginal effects when formulating transport and energy policies, as linear models may fail to capture such complex behavioral responses.

Table 3: Persistent (PE) and transient TE) efficiency (year 2023).

Country	TE	PE
Austria	0.708	0.771
Belgium	0.577	0.742
BosniaHerzegovina	0.817	0.748
Bulgaria	0.630	0.757
Croatia	0.565	0.779
Cyprus	0.731	0.819
Czechia	0.647	0.773
Denmark	0.759	0.778
Estonia	0.657	0.815
Finland	0.658	0.788
France	0.618	0.79
Germany	0.704	0.801
Greece	0.655	0.776
Hungary	0.648	0.765
Iceland	0.462	0.837
Ireland	0.681	0.776
Italy	0.592	0.792
Latvia	0.716	0.788
Lithuania	0.676	0.780
Luxembourg	0.806	0.749
Malta	0.598	0.857
Netherlands	0.743	0.800
Norway	0.731	0.799
Poland	0.528	0.772
Portugal	0.635	0.772
Romania	0.610	0.768
Serbia	0.565	0.762
Slovakia	0.685	0.791
Slovenia	0.705	0.788
Spain	0.675	0.777
Sweden	0.679	0.777
Turkiye	0.686	0.714

Table 3 presents the estimated transient efficiency (TE) in 2023 and persistent efficiency (PE) scores for the 32 countries. The results provide valuable insights into both short-term performance dynamics and long-term structural efficiency across countries. Northern and Western Europe (e.g., Finland, Norway, Netherlands) show high efficiencies reflecting mature institutional frameworks while Southern and Eastern Europe (e.g., Croatia, Romania, Bulgaria) exhibit lower TE and PE, suggesting ongoing structural or productivity challenges.

Overall, the results reveal that persistent efficiency levels (PE) are generally higher than transient efficiency (TE) across most countries. This indicates that while temporary shocks and cyclical fluctuations may affect short-term productivity, the underlying long-term efficiency structures remain relatively stable. On average, the TE score is around 0.68, whereas the PE score is approximately 0.78, suggesting that European economies tend to maintain solid long-term performance even when short-term deviations occur. This pattern is consistent with the theoretical expectation and empirical evidence that institutional quality, sunk infrastructure investments and long-lived capital primarily determine persistent inefficiency, whereas managerial practices and transient shocks determine TE (Filippini and Greene, 2016). The predominance of PE over TE in the cross-section suggests that policy levers aimed at long-term structural improvements (capital renewal, regulatory reform, investment in network quality and intermodal hubs) are likely to generate larger and more durable improvements in aggregate road-transport efficiency than measures that affect only short-run managerial performance.

The empirical findings show that European countries exhibit relatively high levels of persistent efficiency, reflecting strong institutional and structural foundations. However, transient efficiency varies more widely, pointing to differences in adaptability and short-term economic management. Policy efforts should therefore focus on strengthening flexibility and innovation to improve transient efficiency while preserving the stable long-term performance already achieved (Minviel and Sipiläinen, 2021).

5.2. Rebound Effects Estimates

The results of the estimated GMM model for transient efficiency (i.e. SRE) are given in Table 4 while the estimated long-run rebound effect (LRE) in Table 5. The estimated

Table 4: Transient rebound effect via dynamic linear panel model.

	GMM onestep			GMM twostep		
	Estimate	Std.Err	Pvalue	Estimate	Std.Err	Pvalue
fuel cons (Lag1)	0.239	0.122	.	0.372	0.312	
fuel cons (Lag2)	0.014	0.050		0.035	0.102	
efficiency	-0.588	0.063	***	-0.599	-0.082	***
gdp	0.517	0.124	***	0.457	0.192	*
gdp (Lag1)	0.052	0.166		-0.099	-0.281	
roads	0.027	0.010	**	0.032	0.013	*
roads (Lag1)	0.045	0.017	**	0.054	0.039	
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1						

energy efficiency elasticities are -0.6 in the short run and -10.8 in the long run. The long-run rebound estimate indicates that improvements in energy efficiency are associated with a

Table 5: Persistent rebound effect model.

	Estimate	Std.Err	Pvalue
roads	0.039	0.020	.
gdp	0.304	0.037	***
carpass	0.858	0.025	***
diesel price	-0.816	0.349	*
PTE	-10.753	1.003	***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1			

more than proportionate reduction in fuel consumption; specifically, a 1% increase in energy efficiency results in an estimated 10.8% decrease in energy use. This outcome corresponds to what the literature defines as super conservation, wherein energy efficiency gains translate into energy savings exceeding the initial improvement in efficiency.

The smaller long-run rebound estimate is consistent with the expectation that, over time, processes of learning, technological innovation, and knowledge diffusion enable energy end-users to lock in more persistent efficiency gains. In the long run, these mechanisms contribute to sustained behavioral and technological adjustments that reinforce initial energy savings. Moreover, this result may also reflect the influence of growing global awareness and sustained policy efforts under the broader climate change agenda, which have increasingly promoted energy conservation, efficiency standards, and the diffusion of low-carbon technologies.

Figures 3 and 4 illustrate the spatial distribution of the Short-Run Rebound Effect (SRE) associated with transient efficiency and the Long-Run Rebound Effect (LRE) linked to persistent efficiency across 32 European countries in 2023. From Figure 3, it can be observed that the SRE exhibits relatively limited cross-country variation, with estimated values ranging between approximately -0.45 and -0.30. This small heterogeneity suggests that short-run responses to energy efficiency improvements are fairly homogeneous across European countries. Most economies display comparable short-term adjustments, reflecting similar behavioral and technological reactions to efficiency changes. The average point estimate, around -0.4, confirms that energy efficiency improvements generally lead to a proportional reduction in fuel consumption in the short run.

By contrast, Figure 4 reveals a greater degree of heterogeneity in the LRE. The estimated values range from about -9.2 to -8.0, indicating that long-run rebound effects differ more substantially across countries. Northern and Western European countries tend to display smaller rebound magnitudes (closer to -8.0), suggesting that efficiency gains in these economies translate into more persistent and stable energy savings. Conversely, several Southern and Eastern European countries exhibit larger rebound effects (closer to -9.2), reflecting structural differences in energy intensity, technology adoption, and policy maturity.

Overall, the comparison between the SRE and LRE maps highlights two key findings: first, short-run efficiency responses appear relatively homogeneous, consistent with common behavioral adjustments across European transport systems; and second, long-run effects are markedly more differentiated, driven by cross-country disparities in technological capability, policy commitment, and the persistence of efficiency improvements. These spatial patterns emphasize the importance of accounting for heterogeneity in efficiency dynamics when for-

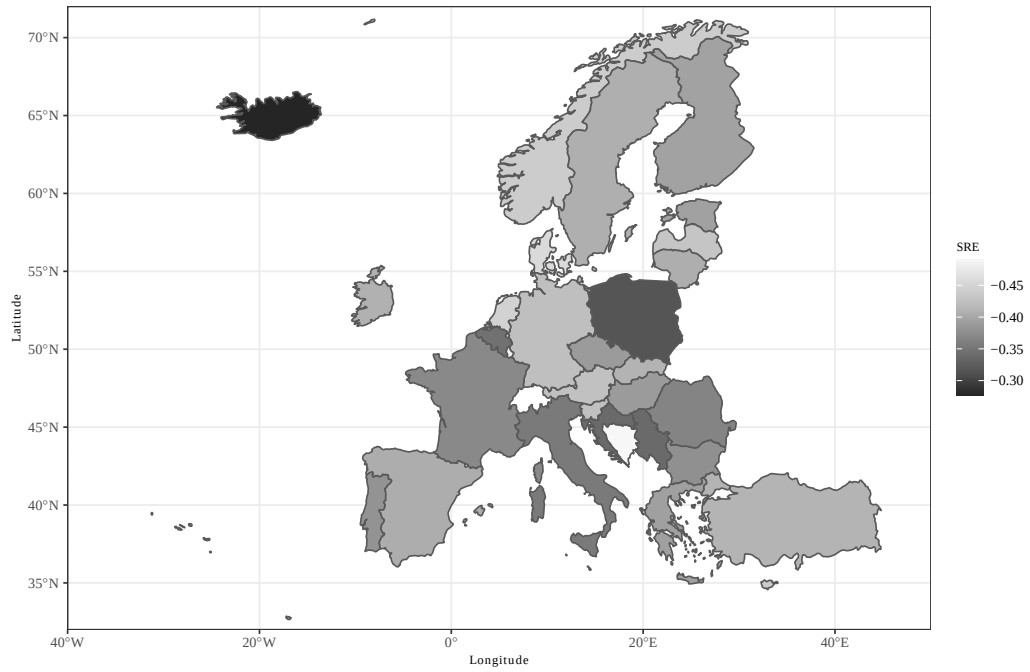


Figure 3: Territorial distribution of pointwise SRE in 2023.

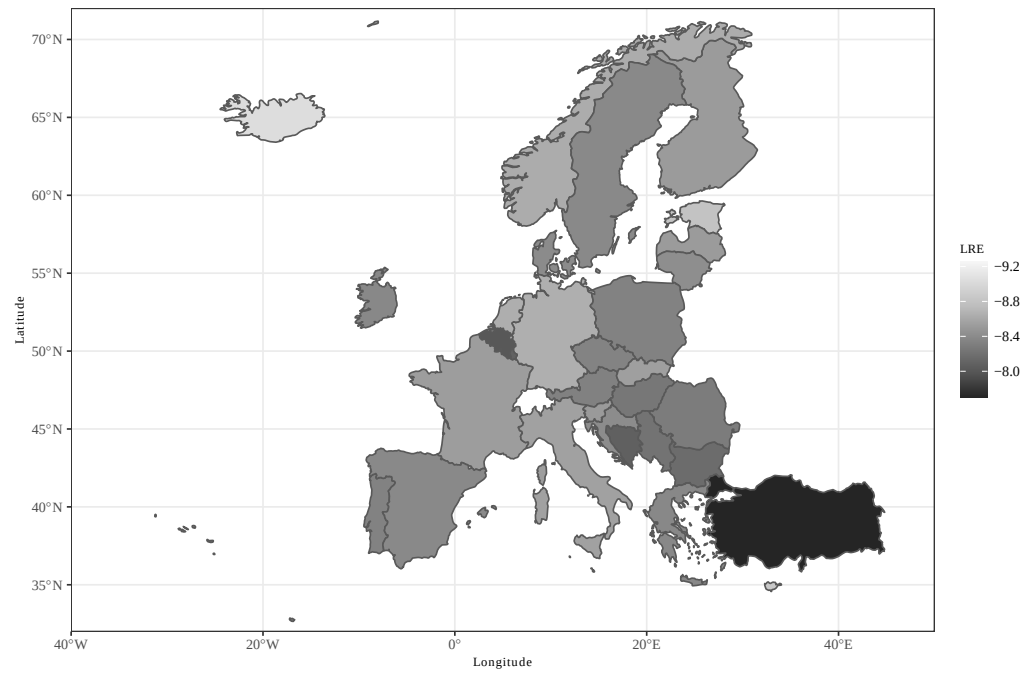


Figure 4: Territorial distribution of pointwise LRE.

mulating long-term energy and climate policies. The results align with previous studies that identify stronger rebound moderation in contexts characterized by innovation, learning, and sustained policy frameworks (Gillingham et al., 2016; Van den Bergh, 2011).

6. Conclusions

Prioritise capital and institutional reforms (modal integration, concession design, emission and fuel-efficiency standards) in countries with low PE to address structural inefficiency.

Use targeted operational programmes (driver training, fleet utilization, real-time logistics) to raise TE in the short run; however, monitor whether transient gains persist or dissipate once shocks pass. Sector studies emphasize that transient gains can be substantial but are often short-lived without supportive structural changes.

Methodologically, the decomposition applied here is consistent with recent advances in dynamic SFA that explicitly control for unobserved heterogeneity while estimating persistent and transient inefficiency components; this reduces the risk of attributing unobserved firm/country heterogeneity to inefficiency and improves identification of true structural (persistent) effects. Applying such frameworks to road transport is important because many component-specific determinants (network design, concession length, capital intensity) are time-invariant or slowly varying, whereas operational performance is volatile. These econometric considerations justify interpreting our PE estimates as reflections of deeper, potentially policy-amenable features of national road-transport systems.

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